9.D Solve Polynomial Equations

Polynomial equations describe relationships where expressions are set equal to zero. Solving these equations requires recognizing algebraic structure and applying appropriate strategies to isolate all possible solutions.

> *"Equations are solved not by guessing, but by revealing their structure."*

Learning Targets

I can solve polynomial equations using factoring.
I can use graphing to approximate solutions.
I can apply the Zero Product Property.
I can verify solutions algebraically and graphically.

Introduction

A polynomial equation is solved by finding all values of x that make the expression equal to zero. These values are called the solutions or zeros of the equation.
Graphically, solutions correspond to the x-intercepts of the function. Algebraically, they are found by rewriting the equation in factored form and applying properties of multiplication.

Key Idea: Solving a polynomial equation means finding all values of x for which the function equals zero.

Connections

Prior Knowledge:
- Factoring expressions and identifying common factors
- Graphing functions and identifying x-intercepts
- Understanding the relationship between equations and their graphs

Cross-Disciplinary Connections:
- Physics: determining when height or velocity equals zero
- Engineering: identifying equilibrium points in systems
- Economics: finding break-even points where profit equals zero

Real-World Applications:
- Determining when a projectile hits the ground
- Finding when revenue equals cost in a business model
- Identifying when a system changes direction or behavior

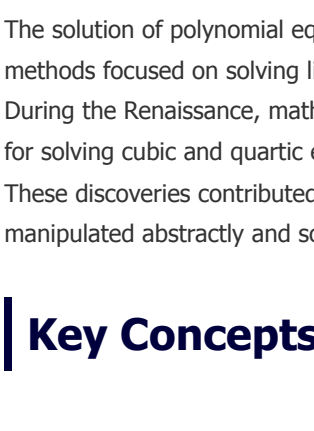
History

The solution of polynomial equations has been studied since ancient civilizations, where early methods focused on solving linear and quadratic equations using arithmetic procedures.
During the Renaissance, mathematicians such as Cardano and Ferrari developed general methods for solving cubic and quartic equations, marking a major advancement in algebra.
These discoveries contributed to the development of symbolic algebra, allowing equations to be manipulated abstractly and solved using systematic methods that are still used today.

Key Concepts

Solutions and Zeros

A solution to a polynomial equation is any value of x that makes the expression equal to zero.
These solutions are also called zeros of the function.
Graphically, solutions correspond to the x-intercepts of the graph.



The Zero Product Property

Zero Product Property:
If $ab = 0$, then $a = 0$ or $b = 0$.

This property allows us to solve equations by factoring.
Example:

$$(x-2)(x+3)=0$$
$$x-2=0 \quad \text{or} \quad x+3=0$$
$$x=2 \quad \text{or} \quad x=-3$$

Factoring to Solve Equations

To solve a polynomial equation, rewrite it in factored form and apply the Zero Product Property.

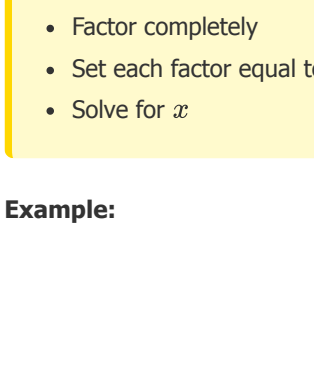
Process:
- Set the equation equal to zero
- Factor completely
- Set each factor equal to zero
- Solve for x

Example:

$$x^2-5x+6=0$$
$$(x-2)(x-3)=0$$
$$x=2,\ x=3$$

Graphing and Solutions

The solutions of a polynomial equation are the x-values where the graph crosses or touches the x-axis.



Key Idea: Algebraic solutions and graphical solutions represent the same values.

Verifying Solutions

Solutions should always be checked by substituting them back into the original equation.
Graphing can also be used to confirm solutions by verifying the x-intercepts.

Vocabulary

Polynomial Equation:
An equation involving a polynomial set equal to zero.

Solution:
A value of x that makes the equation true.

Zero (Root):
A value of x where the function equals zero; corresponds to an x-intercept.

Zero Product Property:
If $ab = 0$, then $a = 0$ or $b = 0$.

Factor:
An expression that divides another expression evenly.

X-intercept:
The point where the graph crosses or touches the x-axis.

Strategies

Solving Polynomial Equations:
1. **Set the equation equal to zero.**
   All solving methods require the equation to be written in standard form.
2. **Factor completely.**
   Use common factoring techniques (GCF, trinomials, grouping, etc.).
3. **Apply the Zero Product Property.**
   Set each factor equal to zero and solve.
4. **Check solutions.**
   Substitute values back into the original equation to verify correctness.
5. **Use graphing when needed.**
   Graph the function to approximate solutions or confirm results.

Examples

Example 1 — Long Division (Cubic)

Use polynomial long division to factor and solve.

$$x^3-4x^2+x+6=0$$

Example 2 — Long Division (Quartic)

Use long division to factor and solve.

$$x^4-5x^3+6x^2+4x-8=0$$

Example 3 — Synthetic Division (Cubic)

Use synthetic division to factor and solve.

$$x^3-3x^2-4x+12=0$$

Example 4 — Synthetic Division (Quartic)

Use synthetic division to factor and solve.

$$x^4-2x^3-7x^2+8x+12=0$$

Example 5 — Box Method (Quartic)

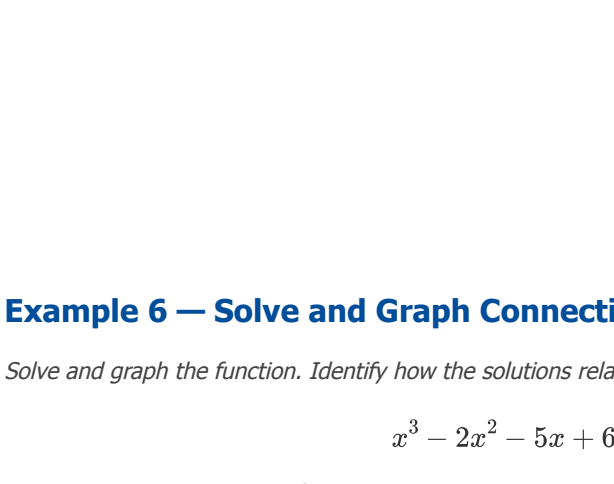
Use the box method to factor and solve.

$$x^4+2x^3-x^2-2x=0$$

Example 6 — Solve and Graph Connection

Solve and graph the function. Identify how the solutions relate to the graph.

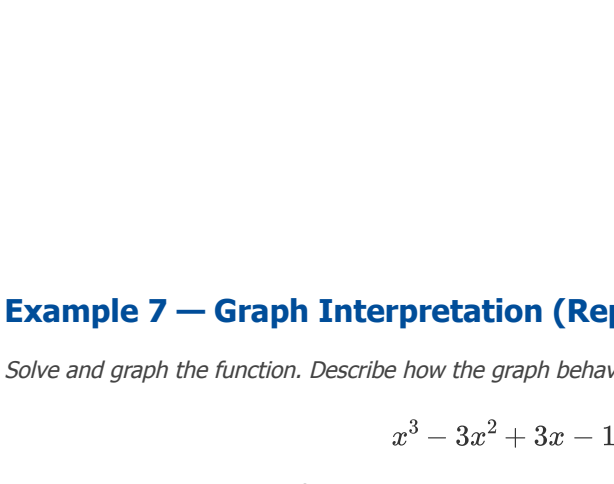
$$x^3-2x^2-5x+6=0$$



Example 7 — Graph Interpretation (Repeated Root)

Solve and graph the function. Describe how the graph behaves at each intercept.

$$x^3-3x^2+3x-1=0$$



Practice Set

Level 1 — Easy

1. Solve the equation.

$$x^2-5x+6=0$$

2. Solve the equation.

$$x^2-9=0$$

3. Solve the equation.

$$x^3-4x^2=0$$

4. Solve the equation.

$$x^2+x-6=0$$

5. Solve the equation.

$$x^3-x^2-6x=0$$

Level 2 — Medium

6. Factor and solve.

$$x^3-3x^2-4x=0$$

7. Factor and solve.

$$x^4-4x^2=0$$

8. Use grouping to solve.

$$x^3-x^2-4x+4=0$$

9. Solve the equation.

$$x^4-x^3-6x^2+6x=0$$

10. Solve the equation.

$$x^3+2x^2-x-2=0$$

Level 3 — Hard

11. Use synthetic division to factor and solve.

$$x^3-2x^2-5x+6=0$$

12. Use synthetic division to factor and solve.

$$x^3-4x^2+x+6=0$$

13. Use division to factor and solve.

$$x^4-5x^3+6x^2+4x-8=0$$

14. Use synthetic division to factor and solve.

$$x^4-2x^3-7x^2+8x+12=0$$

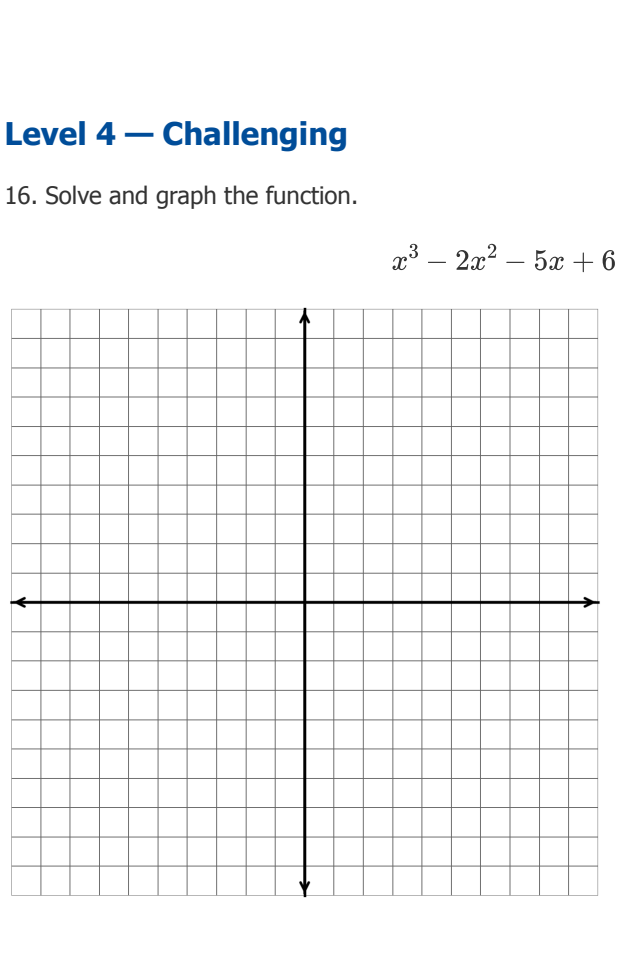
15. Solve the equation.

$$x^4+2x^3-x^2-2x=0$$

Level 4 — Challenging

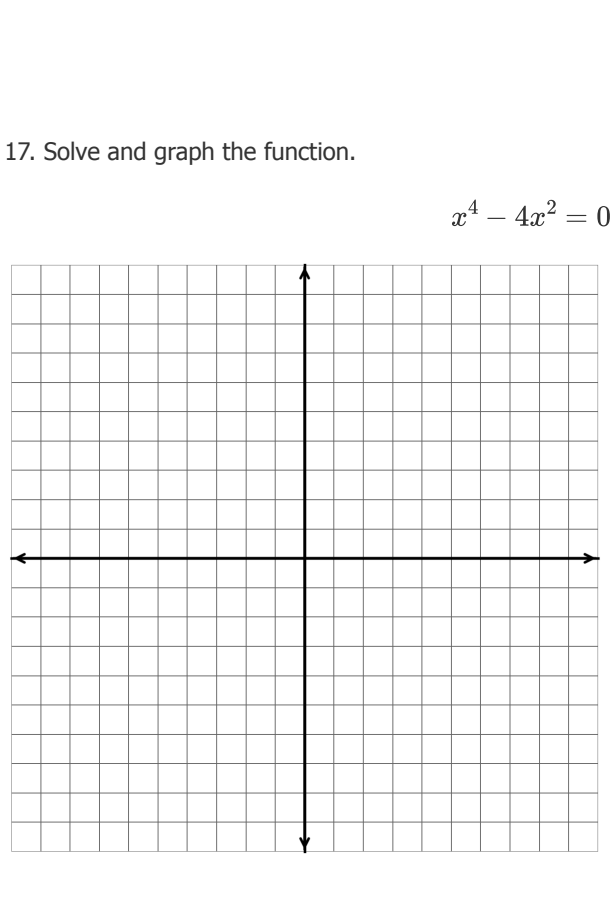
16. Solve and graph the function.

$$x^3-2x^2-5x+6=0$$



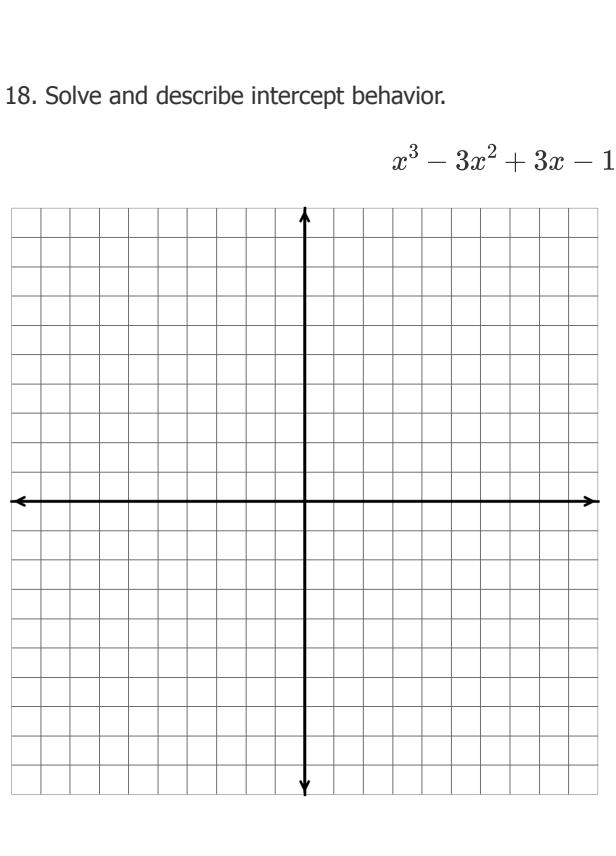
17. Solve and graph the function.

$$x^4-4x^2=0$$



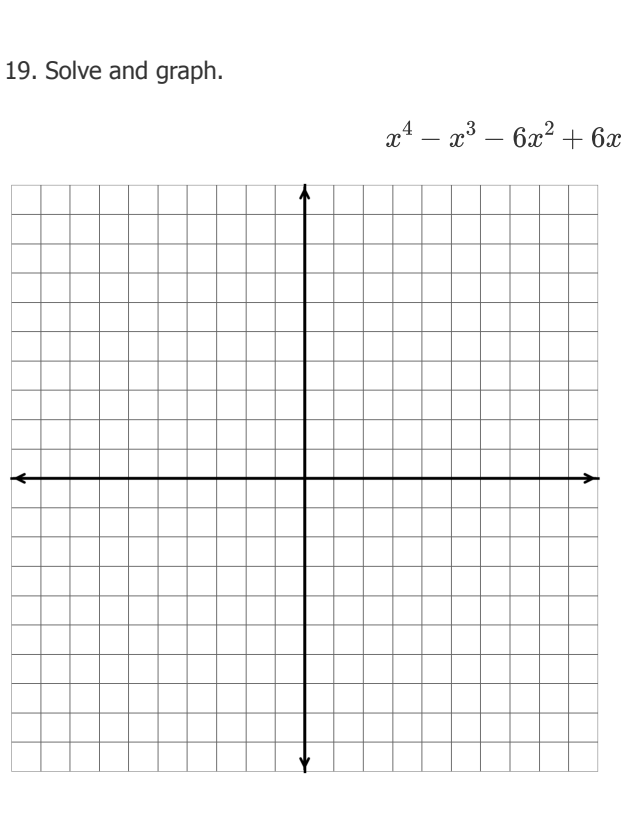
18. Solve and describe intercept behavior.

$$x^3-3x^2+3x-1=0$$



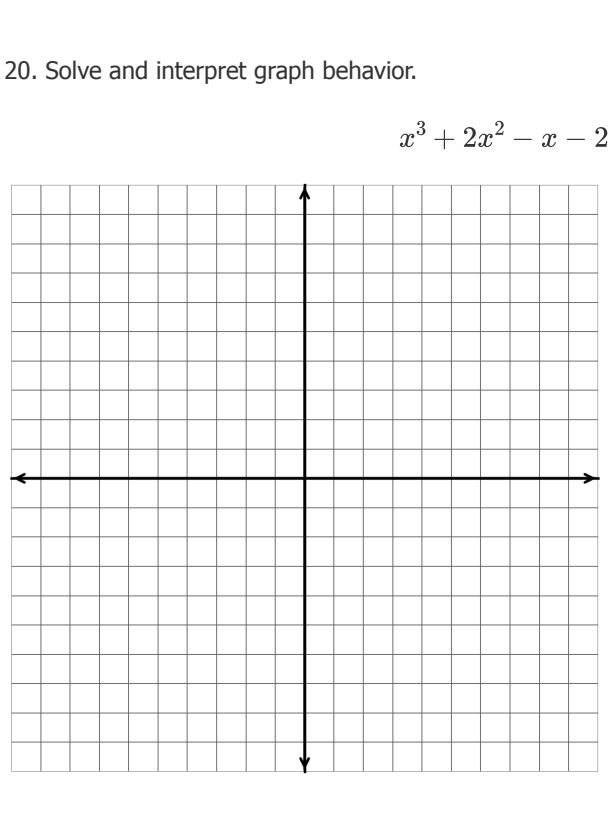
19. Solve and graph.

$$x^4-x^3-6x^2+6x=0$$



20. Solve and interpret graph behavior.

$$x^3+2x^2-x-2=0$$



Level 5 — Justification / Conceptual

21. Explain why factoring is required before applying the Zero Product Property.

22. Explain how solutions relate to x-intercepts.

23. Justify why synthetic division can simplify solving higher-degree polynomials.

24. Explain how repeated roots affect the graph of a polynomial.

25. Describe how algebraic solutions and graphs represent the same information.

Answer Key

Level 1 — Easy

1. $x=2,3$
2. $x=3,-3$
3. $x=0,4$
4. $x=2,-3$
5. $x=0,3,-2$

Level 2 — Medium

6. $x=0,4,-1$
7. $x=0,2,-2$
8. $x=1,-2,2$
9. $x=0,1,2,-3$
10. $x=1,-2,1$

Level 3 — Hard

11. $x=1,3,-2$
12. $x=2,3,-1$
13. $x=2,1,-2,4$
14. $x=3,2,-2,-1$
15. $x=0,-1,1,-2$

Level 4 — Challenging

16. $x=1,3,-2$ (crosses at each root)
17. $x=0,2,-2$ (touches at 0, crosses at ±2)
18. $x=1$ (multiplicity 3, flattens at intercept)
19. $x=0,1,2,-3$
20. $x=1,-2,1$ (double root at 1)

Level 5 — Justification

21–25. Answers vary (conceptual)

Detailed Example Solutions

Example 1 — Factoring a Quadratic

$$x^2-5x+6=0$$

Factor:

$$(x-2)(x-3)=0$$

Apply Zero Product Property:

$$x=2,\ x=3$$

Example 2 — Long Division (Quartic)

$$x^4-5x^3+6x^2+4x-8=0$$

Divide by $(x-2)$ using long division to reduce the polynomial.
Resulting factorization allows solving remaining polynomial.

Example 3 — Synthetic Division (Cubic)

$$x^3-3x^2-4x+12=0$$

Use synthetic division with root $x=3$:
Factorization leads to:

$$(x-3)(x+2)(x-2)=0$$

Solutions:

$$x=3,\ x=-2,\ x=2$$

Example 4 — Synthetic Division (Quartic)

Use synthetic division repeatedly to reduce the polynomial into linear factors.

Example 5 — Box Method

$$x^4+2x^3-x^2-2x=0$$

Group and factor:

$$x(x^3+2x^2-x-2)$$

Then factor remaining cubic using grouping or division.

Example 6 — Solve and Graph Connection

$$x^3-2x^2-5x+6=0$$

Factor using synthetic division:

$$(x-1)(x-3)(x+2)=0$$

Solutions correspond to x-intercepts at $x=1,3,-2$.
The graph crosses the x-axis at each intercept.

Example 7 — Repeated Root Behavior

$$x^3-3x^2+3x-1=0$$

Factor:

$$(x-1)^3=0$$

Solution:

$$x=1$$

The graph touches and flattens at $x=1$ due to multiplicity 3.